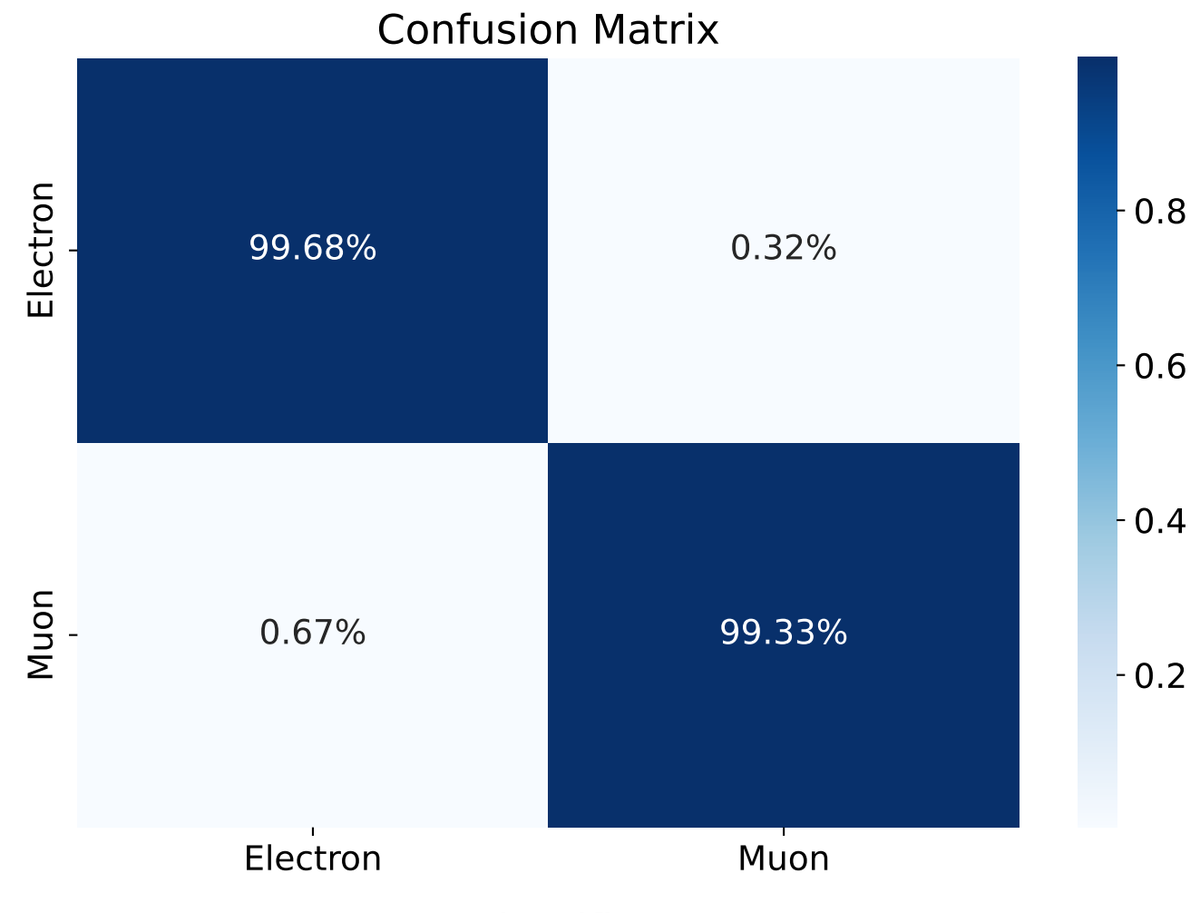
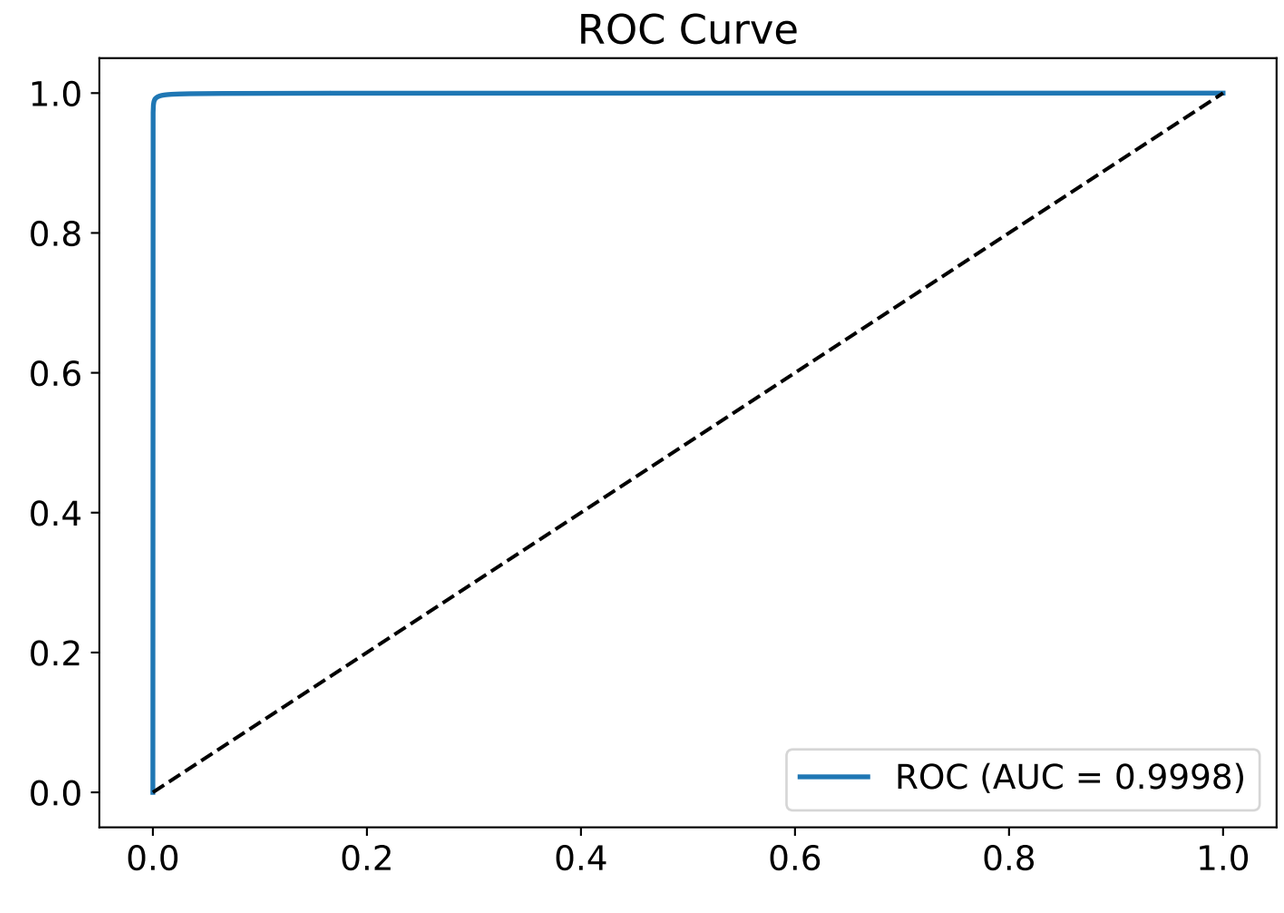
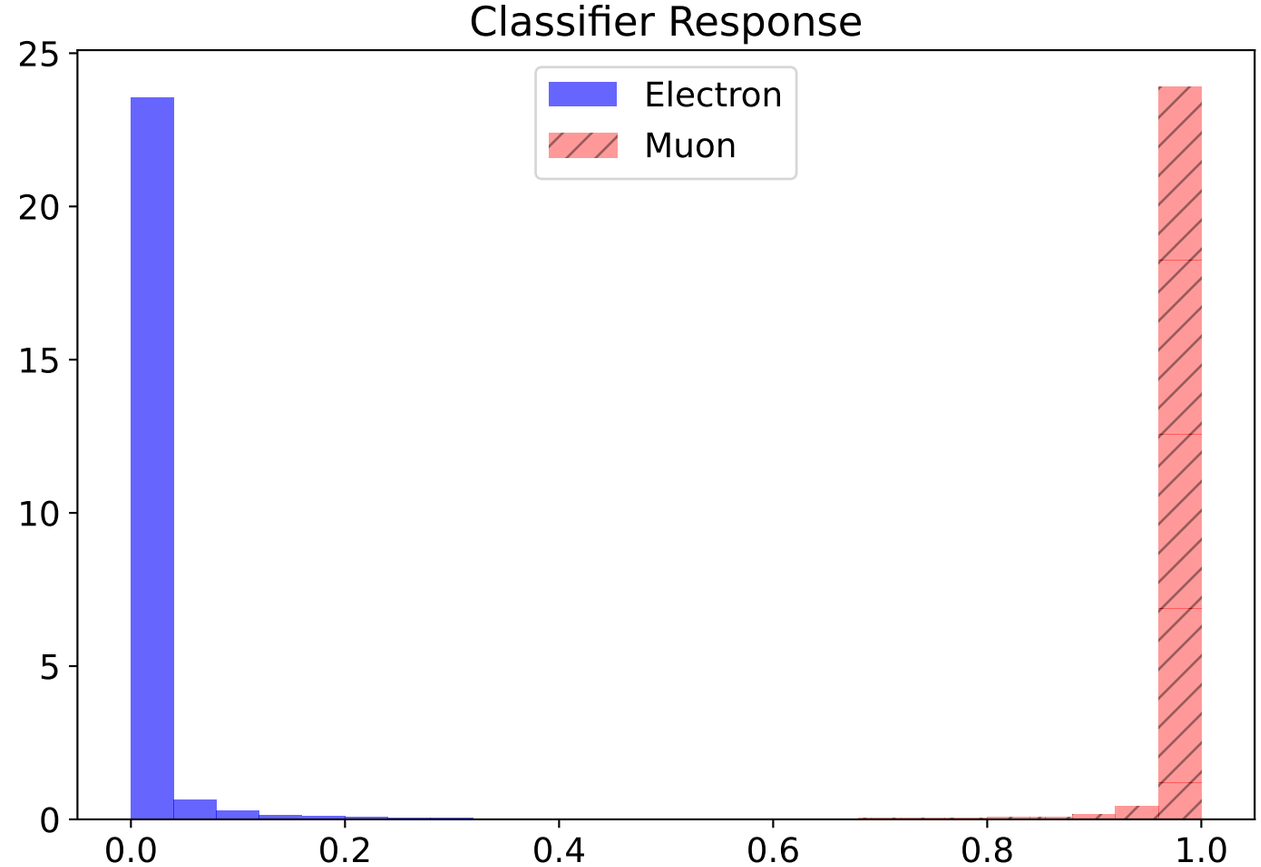
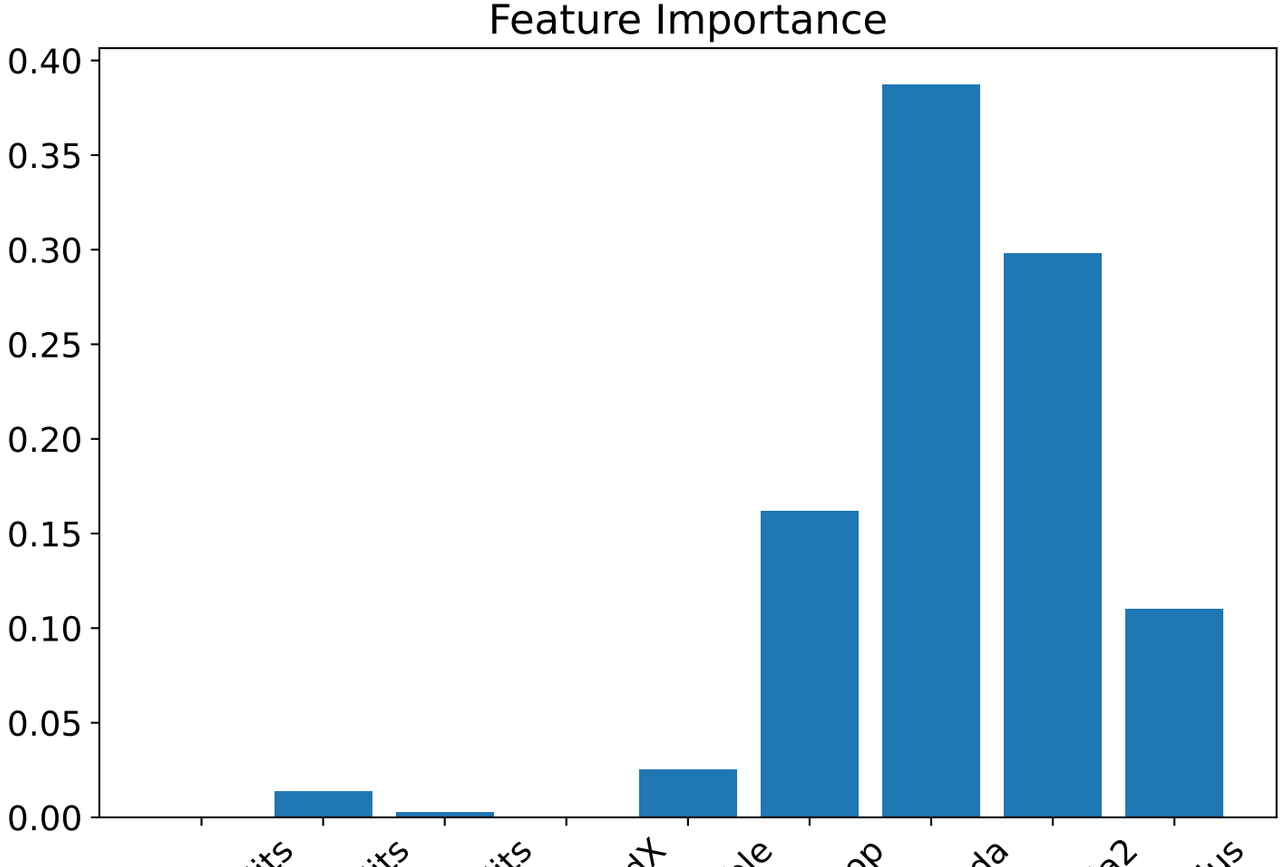
EtaRange = FullRange (TEST DATA) - Accuracy: 0.994

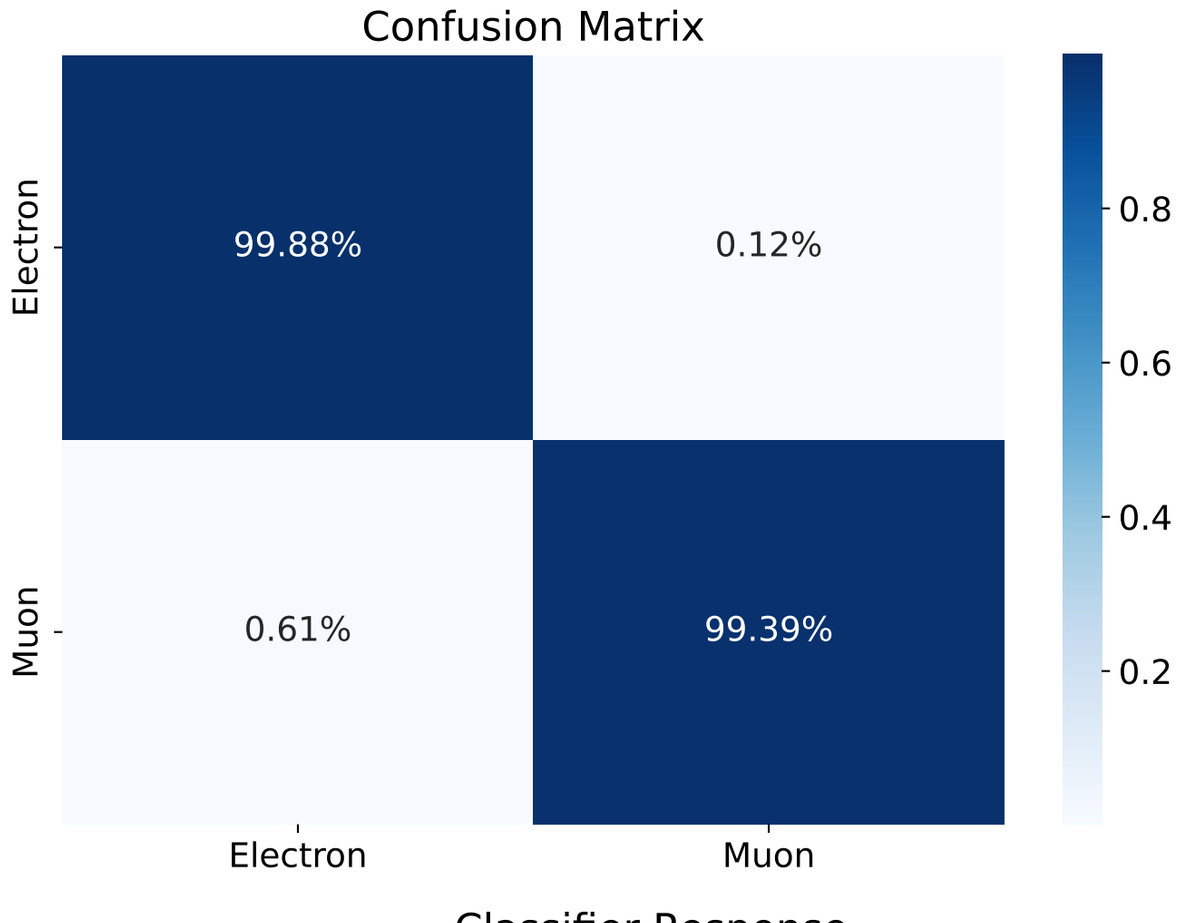


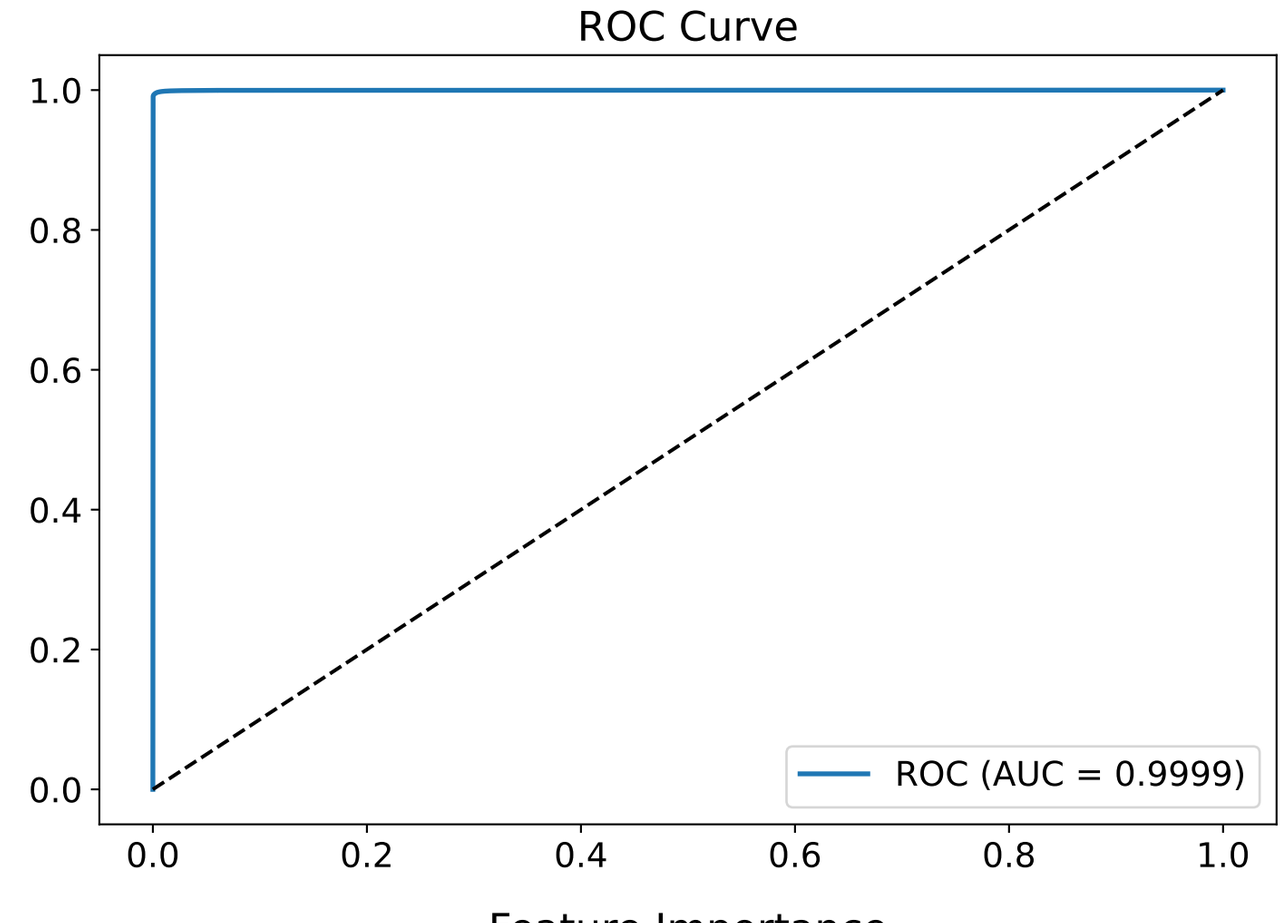


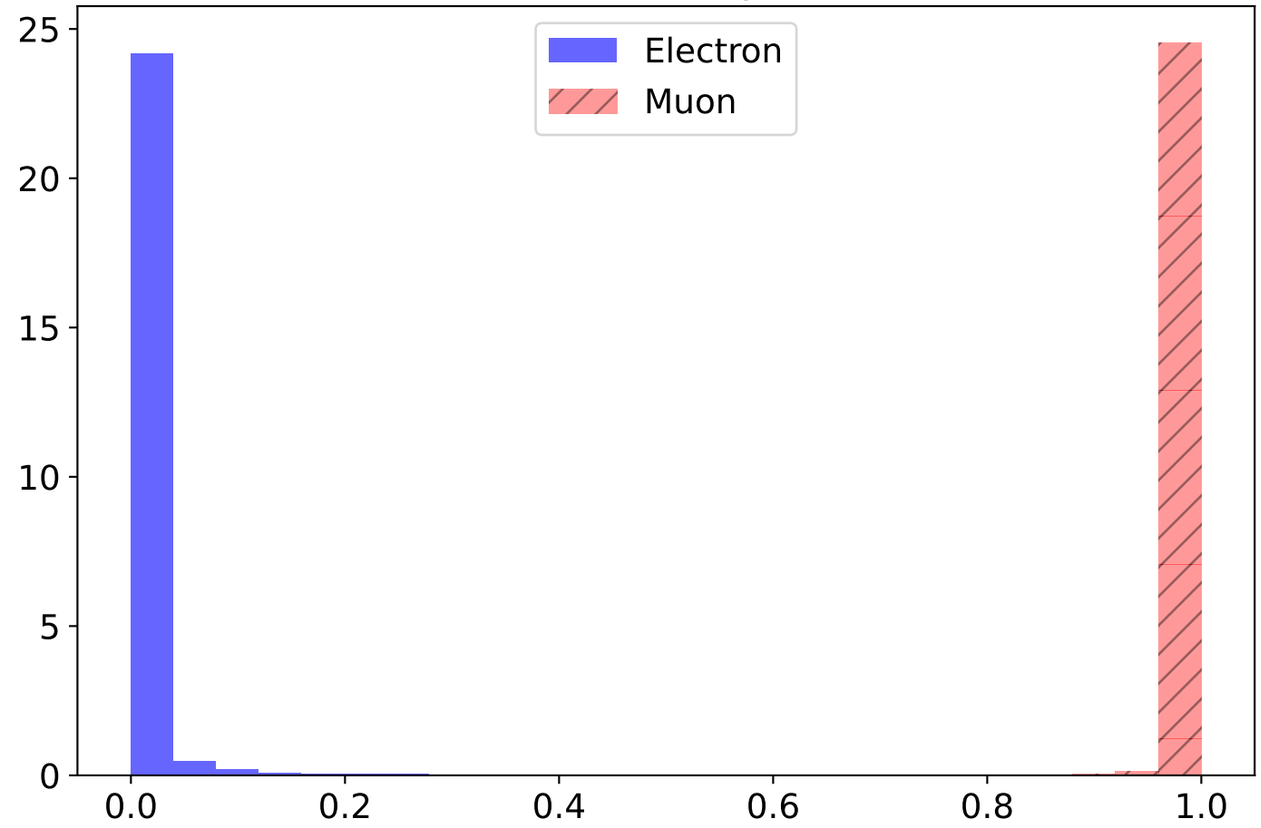


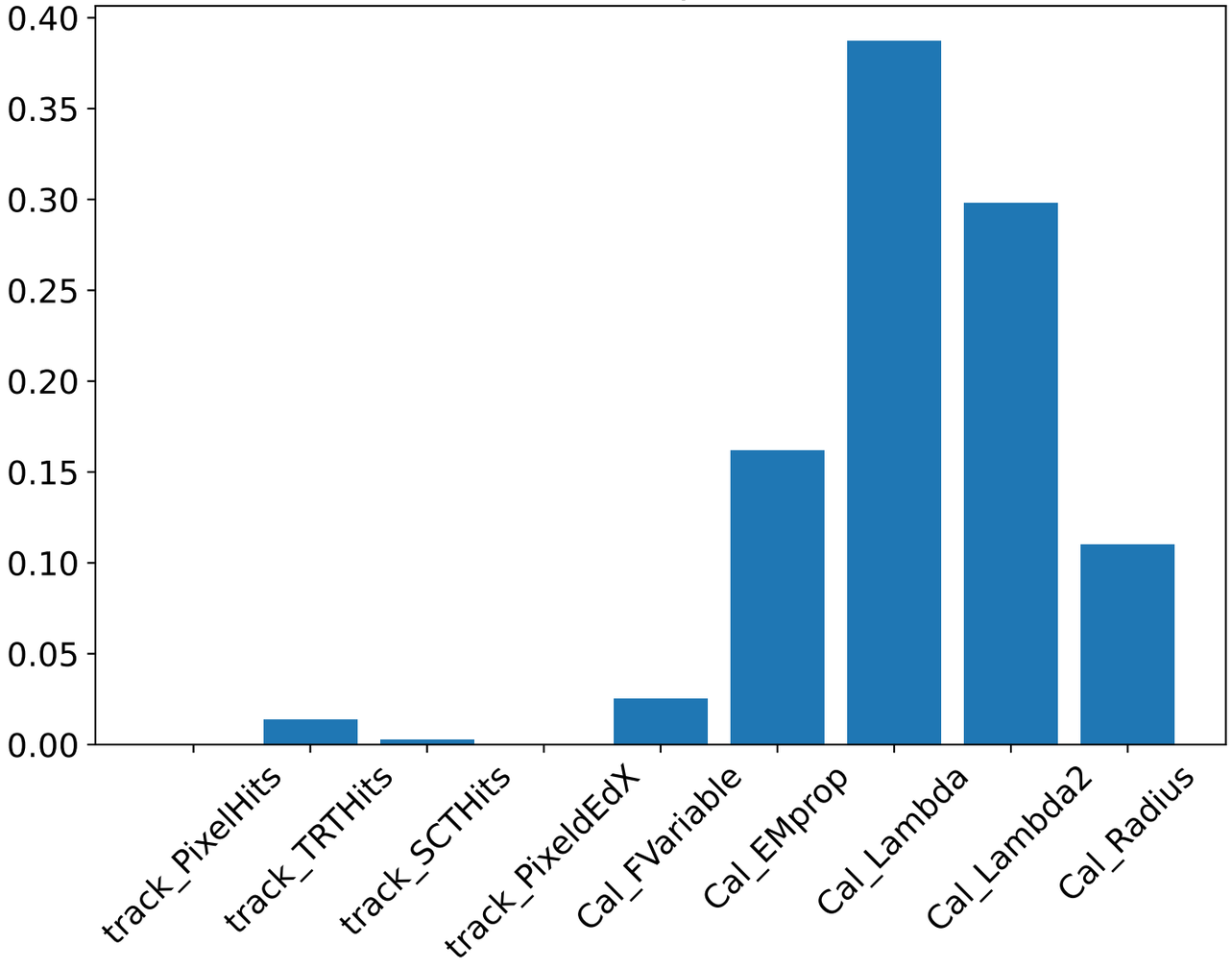


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.995

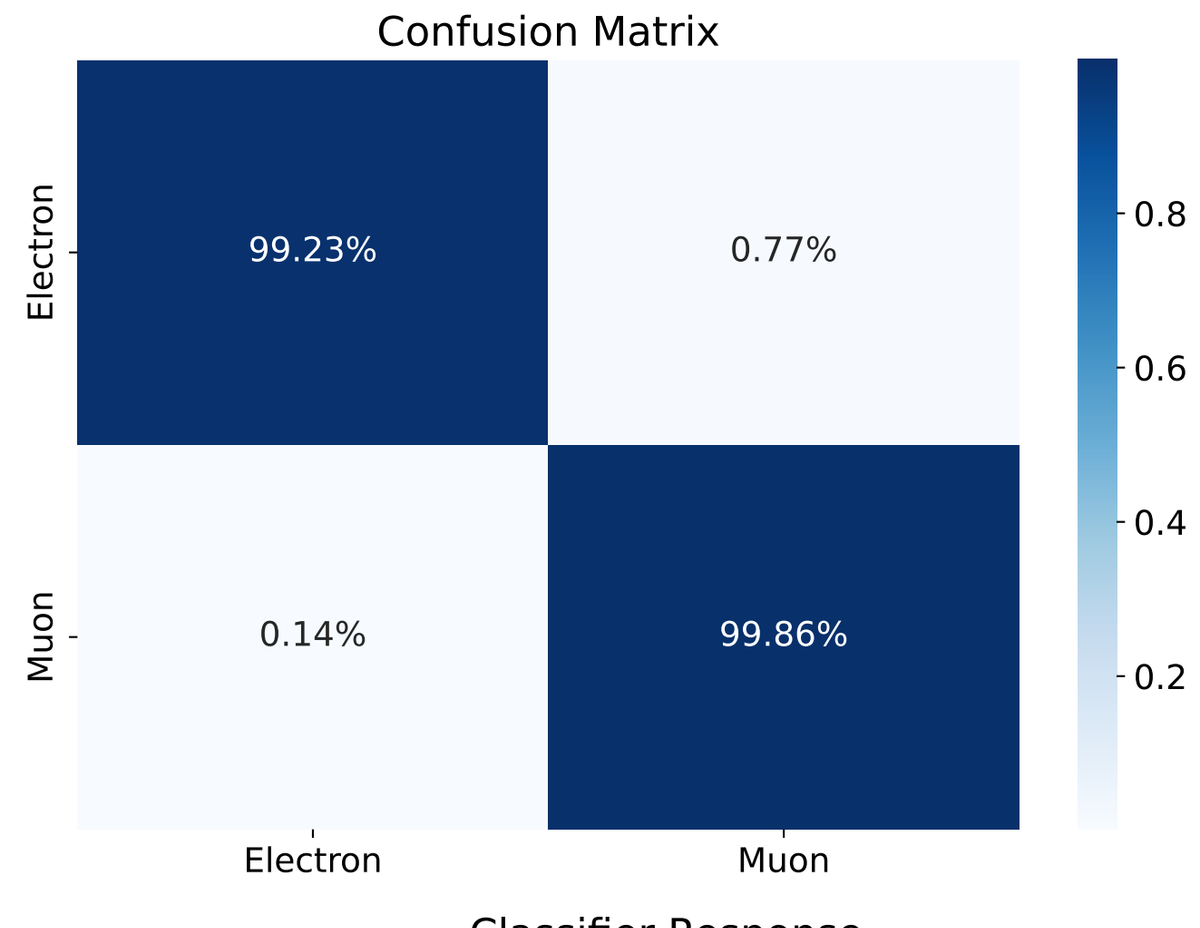


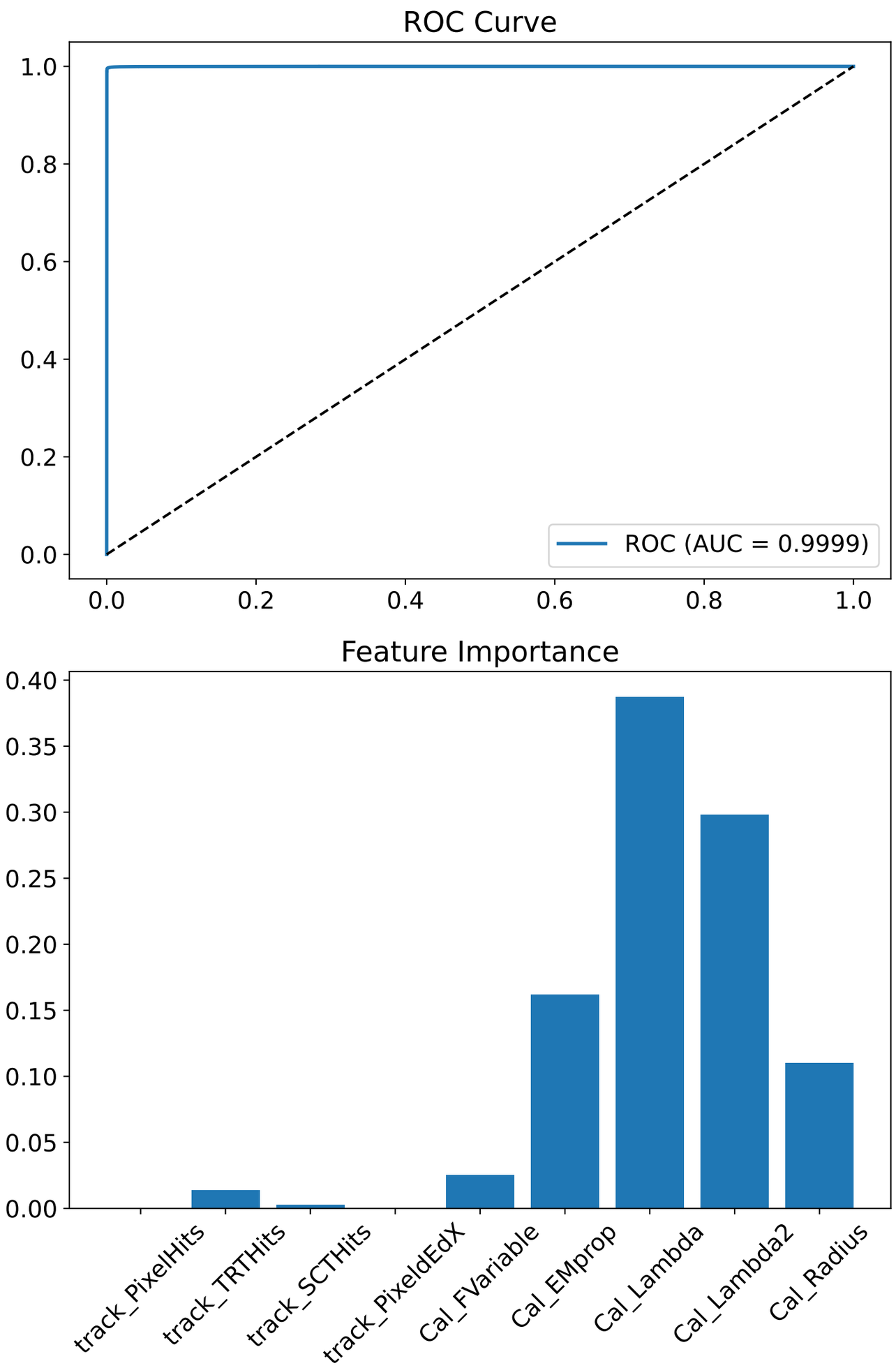




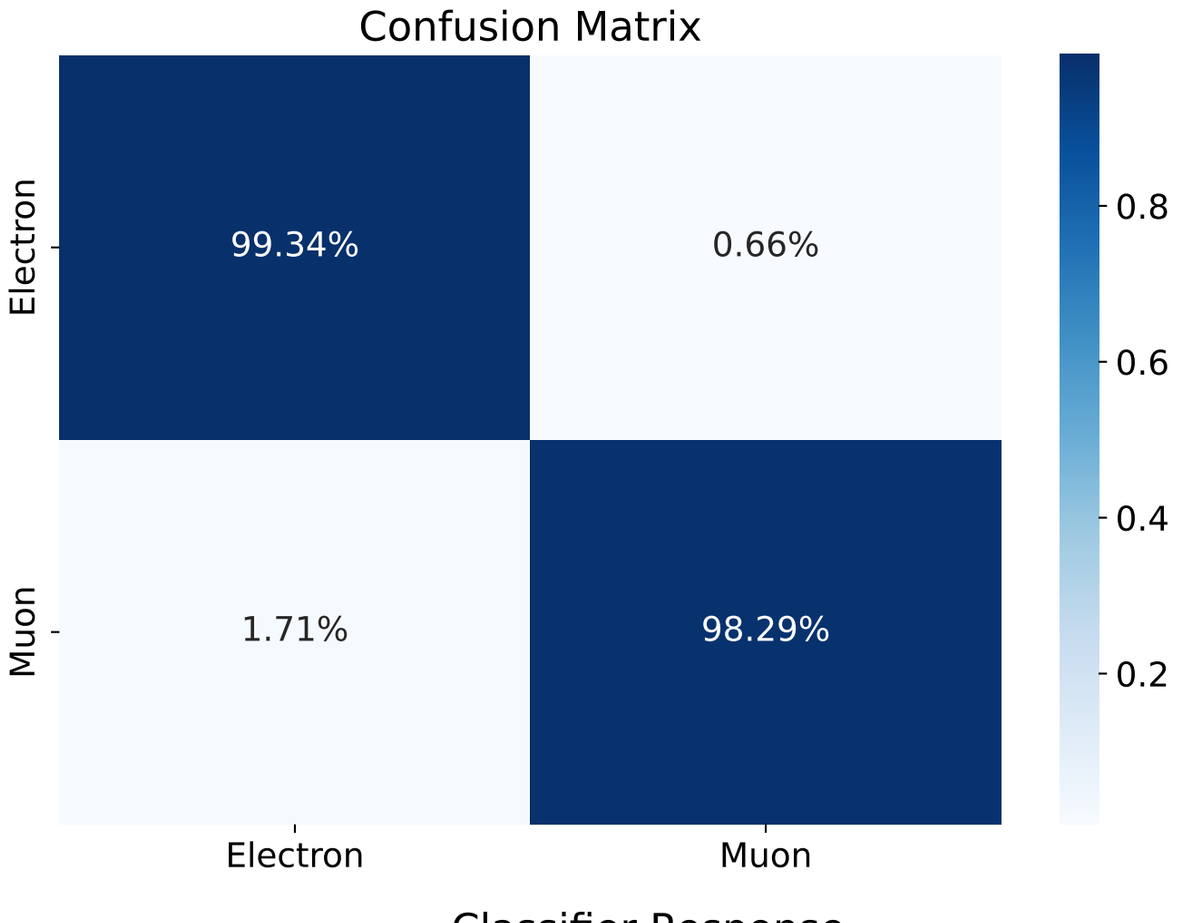


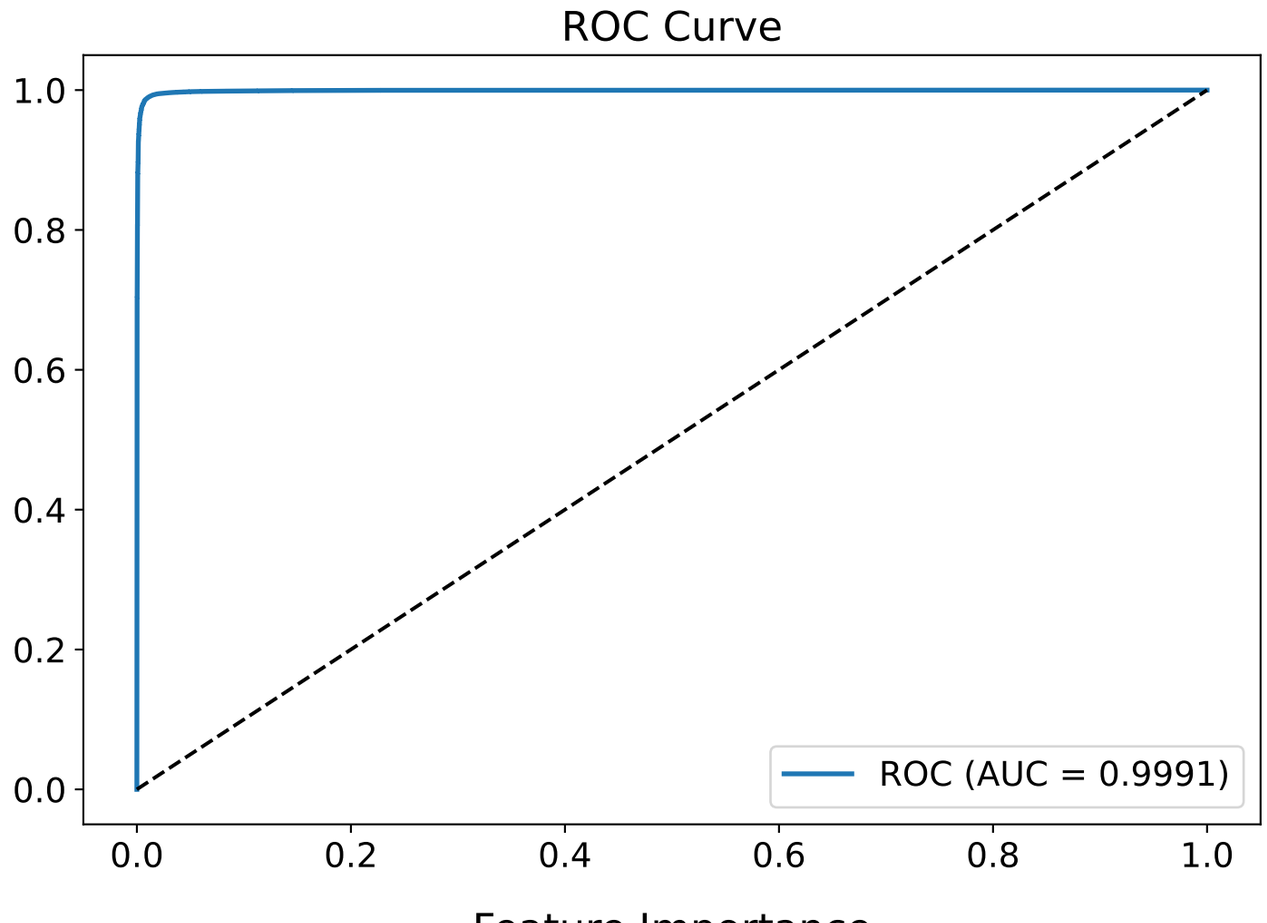
EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.997

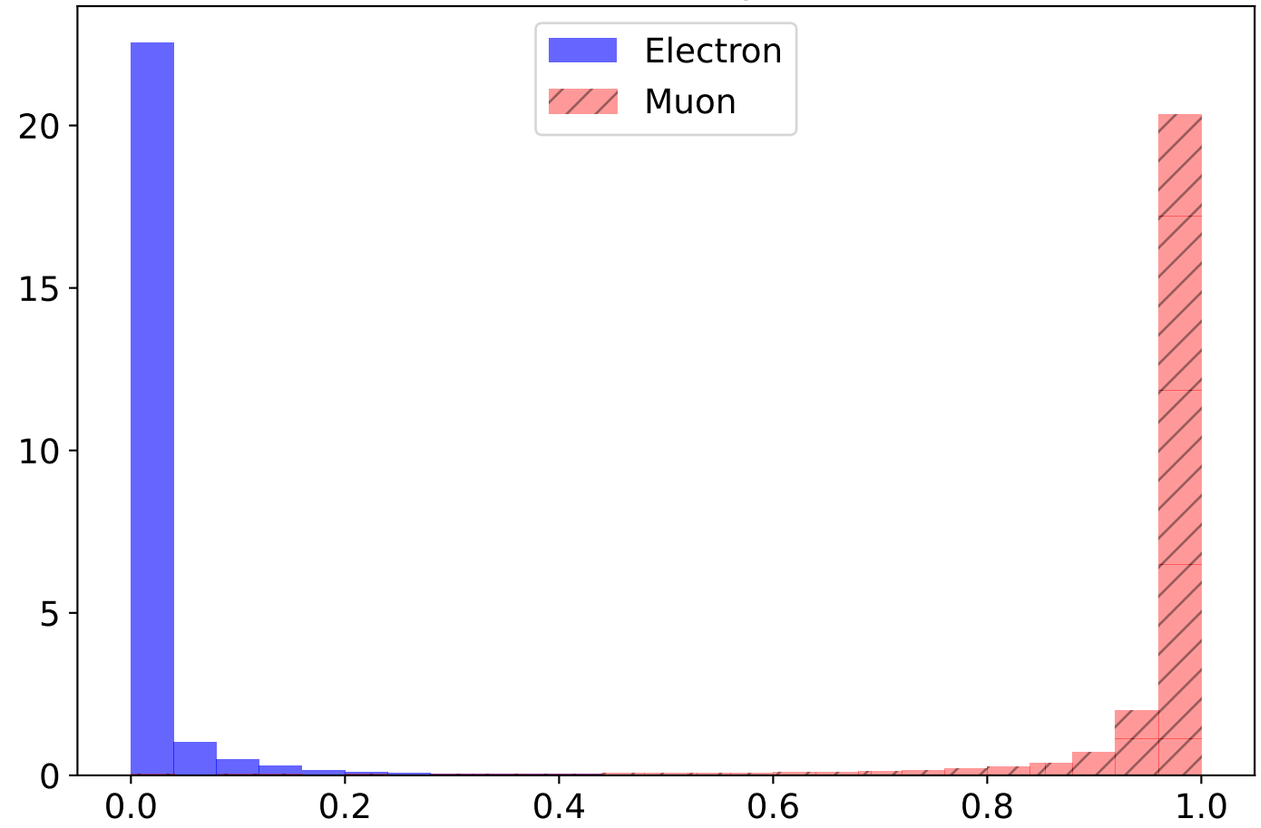


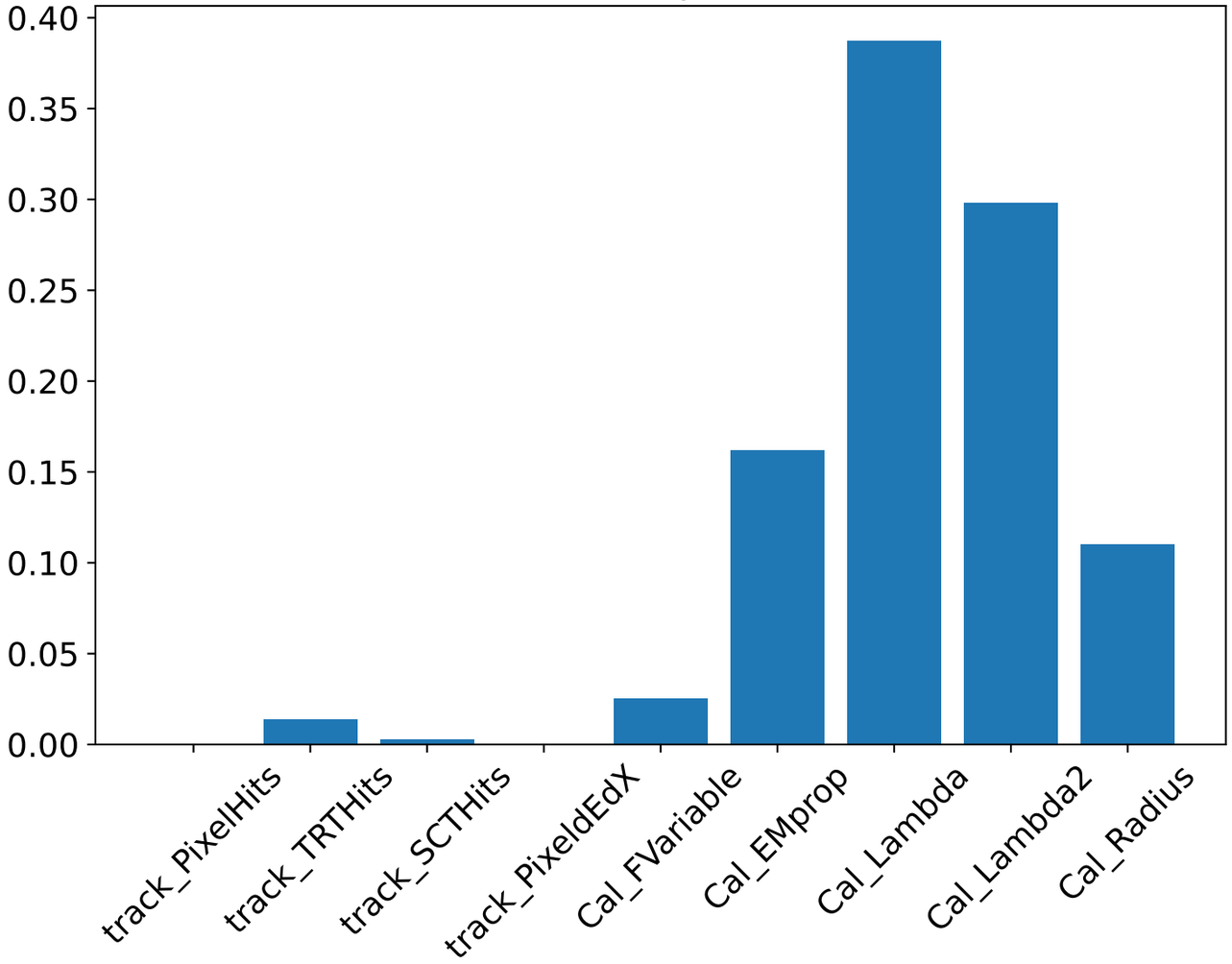


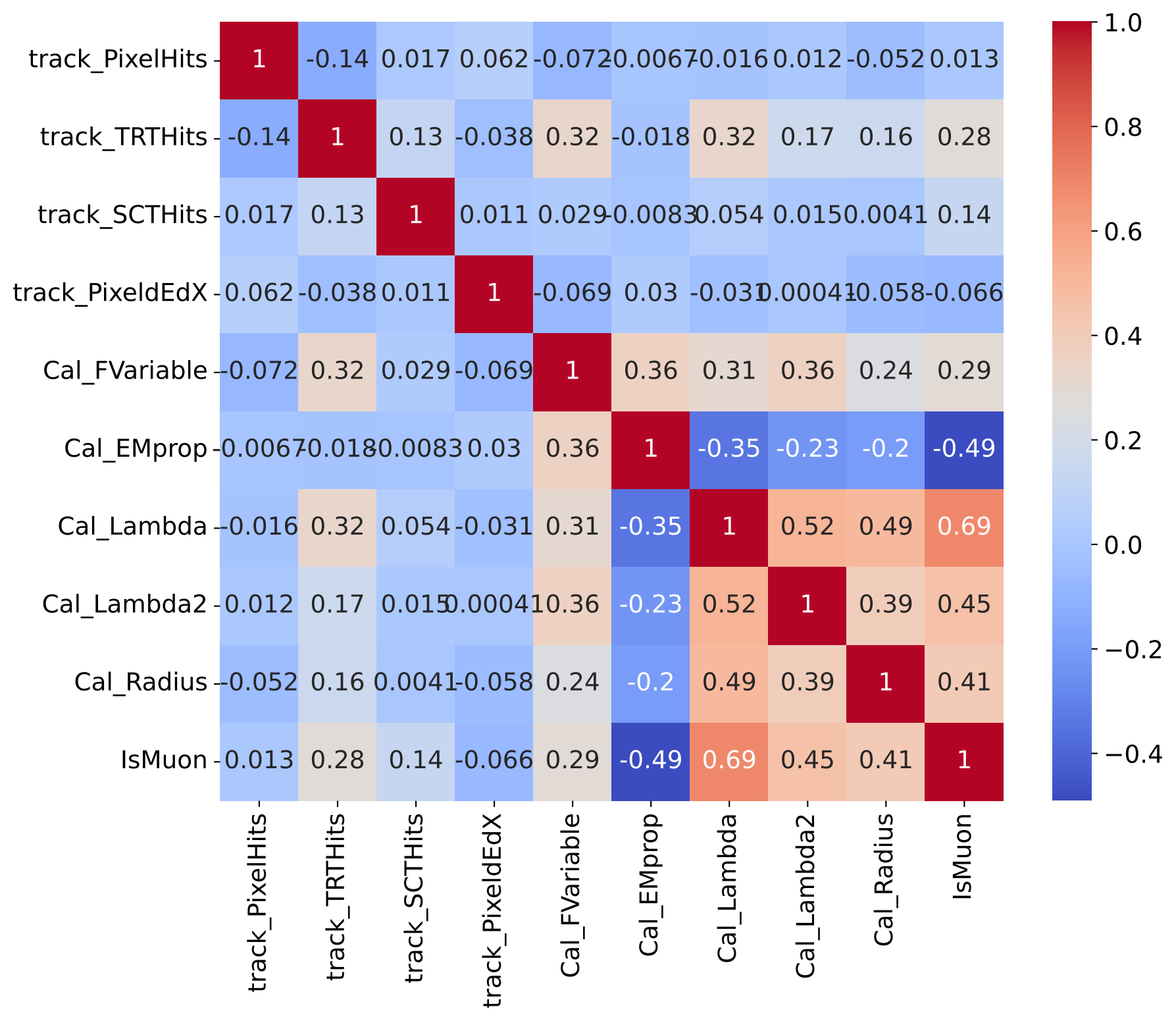
EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.985










track_PixelHits 1 -0.14 0.017 0.062 -0.072 0.0067 -0.016 0.012 -0.052 0.013
track_TRTHits -0.14 1 0.13 -0.038 0.32 -0.018 0.32 0.17 0.16 0.28
track_SCTHits 0.017 0.13 1 0.011 0.029 -0.0083 0.054 0.015 0.0041 0.14
track_PixeldEdX 0.062 -0.038 0.011 1 -0.069 0.03 -0.031 0.00041 0.058 -0.066
Cal_FVariable -0.072 0.32 0.029 -0.069 1 0.36 0.31 0.36 0.24 0.29
Cal_EMprop 0.0067 -0.018 -0.0083 0.03 0.36 1 -0.35 -0.23 -0.2 -0.49
Cal_Lambda -0.016 0.32 0.054 -0.031 0.31 -0.35 1 0.52 0.49 0.69
Cal_Lambda2 0.012 0.17 0.015 0.00041 0.36 -0.23 0.52 1 0.39 0.45
Cal_Radius -0.052 0.16 0.0041 -0.058 0.24 -0.2 0.49 0.39 1 0.41
IsMuon 0.013 0.28 0.14 -0.066 0.29 -0.49 0.69 0.45 0.41 1
track_PixelHits
track_TRTHits
track_SCTHits
track_PixeldEdX
Cal_FVariable
Cal_EMprop
Cal_Lambda
Cal_Lambda2
Cal_Radius
IsMuon
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Classification Reports Summary
Trained: MCData.root | Test: MCData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	158875
Muon	1.00	0.99	1.00	497767
accuracy			0.99	656642
macro avg	0.99	1.00	0.99	656642
weighted avg	0.99	0.99	0.99	656642

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	106862
Muon	1.00	0.99	1.00	290779
accuracy			1.00	397641
macro avg	0.99	1.00	0.99	397641
weighted avg	1.00	1.00	1.00	397641

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	31399
Muon	1.00	1.00	1.00	125372
accuracy			1.00	156771
macro avg	1.00	1.00	1.00	156771
weighted avg	1.00	1.00	1.00	156771

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.94	0.99	0.96	20614
Muon	1.00	0.98	0.99	81616
accuracy			0.98	102230
macro avg	0.97	0.99	0.98	102230
weighted avg	0.99	0.98	0.99	102230